

COMPACT PERMANENT MAGNETS FOR SMALL BORE ACCELERATORS

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Abstract

In synchrotron light sources, electromagnets are used to bend and accelerate electron beams. In 4th generation sources, the electron beam can fit in smaller bore accelerators, allowing the use of permanent magnets, which have many advantages over electromagnets. This paper focuses on the mechanical design, fabrication and testing of two compact permanent magnet systems. The first is a dipole-quadrupole magnet assembly, providing a 1 T dipole + 50 T/m quadrupole field to steer the electron. The second is an assembly to adjust a set of tuner and corrector permanent magnets. Regardless of the magnetic forces involved, the tuner magnets can rotate simultaneously and provide a ± 1 T/m quadrupole field for in situ quadrupole focusing strength adjustments, while corrector magnets can be oriented into a prescribed configuration to compensate for small field errors. Prototypes for the dipole-quadrupole, tuner and corrector holders were manufactured and tested, validating the conceptual design.

INTRODUCTION

The Berkeley Center for Magnet Technology (BCMT) group at Lawrence Berkeley National Laboratory (LBNL) is investigating permanent magnets as an alternative to commonly used electromagnets for small bore accelerator lattices, like 4th generation light sources.

Unlike electromagnets, permanent magnet alternatives are significantly smaller and do not require electricity nor a cooling system to operate. Choosing permanent magnets as an alternative comes with challenges though. First, it is known that the strength of a permanent magnet can be affected by environmental changes, like high temperatures and radiation exposure [1]. Second, the assembly process of rare earth magnets can be difficult since they strongly interact with ferromagnetic objects in proximity. Third, magnetic field errors introduced during the fabrication process may be difficult to correct.

Dipole and quadrupole permanent magnet systems have been studied as alternatives to electromagnets at other accelerator facilities [2, 3], where this paper focuses on the mechanical design, fabrication and testing of a stronger compact permanent magnet system: comprised of a combined function dipole-quadrupole (DQ) and a corrector-tuner (CT) assembly. The latter two challenges of permanent magnets mentioned above are the focus this paper will address.

DIPOLE-QUADRUPOLE

Mechanical Design

A combined function DQ assembly is comprised of eleven wedge shape neodymium rare earth permanent magnets, designed to provide the magnetic strength required to steer the electrons. Each magnet is magnetized across the width of the wedge, and the magnetic arrangement resembles a Halbach array (Fig. 1). Additionally, the field orientation and strength of each magnet is verified with a Helmholtz Coil [4] before assembly.

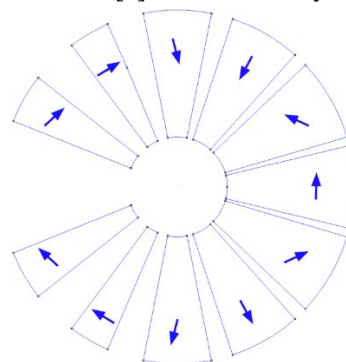


Figure 1: Individual magnetic field direction for DQ assembly.

The functional features of a DQ assembly are shown in Fig. 2. The set screws and guard plates secure the magnets within their respective pockets, whereas the hinge is designed to open and close the assembly to be easily inserted or removed from around a vacuum chamber.

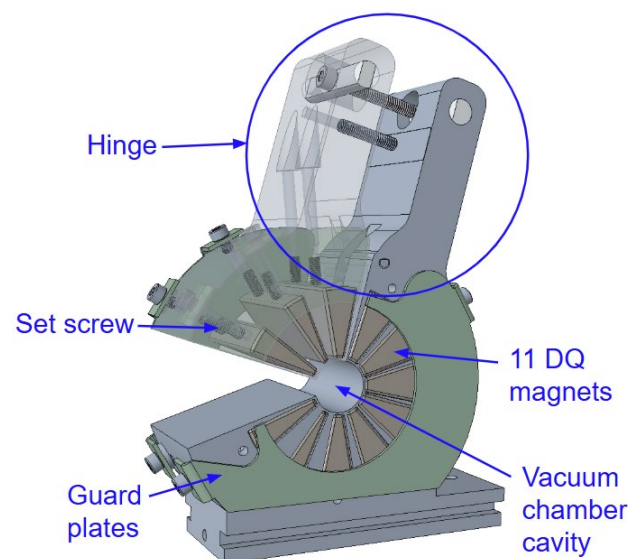


Figure 2: CAD model of DQ assembly showing functional features.

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Fabrication

The DQ assembly utilizes hybrid manufacturing techniques (Fig. 3). The two main body parts encapsulating the DQ magnets are 3D printed from AlSi10Mg, then post processed on a Computer Numerical Control (CNC) machine. The hinge pins are machined from 18-8 SS and the guard plates from 5052 Al.

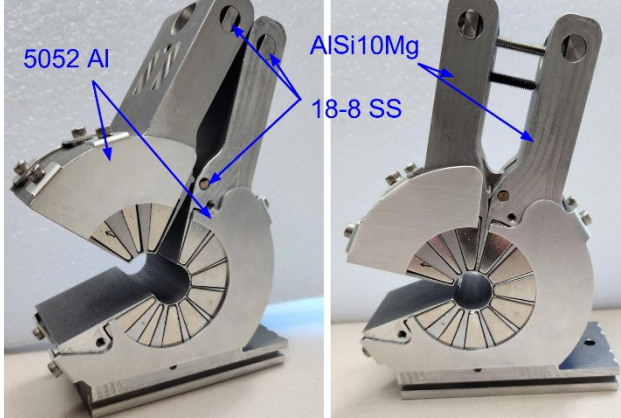


Figure 3: DQ assembly opened (left) and closed (right).

Magnetic Performance

The measured magnetic performance for the DQ assembly is presented on Fig. 4 below, along with the stretch wire measurement system.

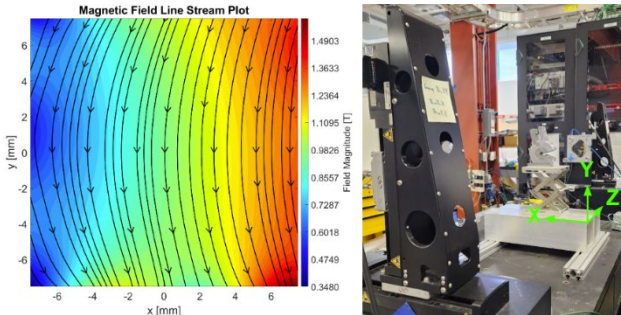


Figure 4: DQ assembly measured magnetic field stream plot (left) and stretch wire measurement setup (right).

The magnetic field line stream plot is determined from equation (1) below [5].

$$B_y(z) + iB_x(z) = \sum_{n=1}^{\infty} (B_n + iA_n) \left(\frac{x + iy}{R} \right)^{n-1} \quad (1)$$

Note that R is the reference radius used on the stretched wire measurement system. The magnetic field multipole component requirements are ≥ 0.78 T dipole + 50 T/m quadrupole. The magnetic field measured components are 0.96 T dipole + 49.7 T/m quadrupole, achieving set requirements.

CORRECTOR-TUNER

An arrangement of tuner magnets and an arrangement of corrector magnets, each with a specific holder, are combined into one CT assembly.

Mechanical Design

The tuner assembly is a set of eight diametrically magnetized neodymium rare earth permanent magnets arranged to form a quadrupole field that can be adjusted from full to zero strength quadrupole to focus the electrons.

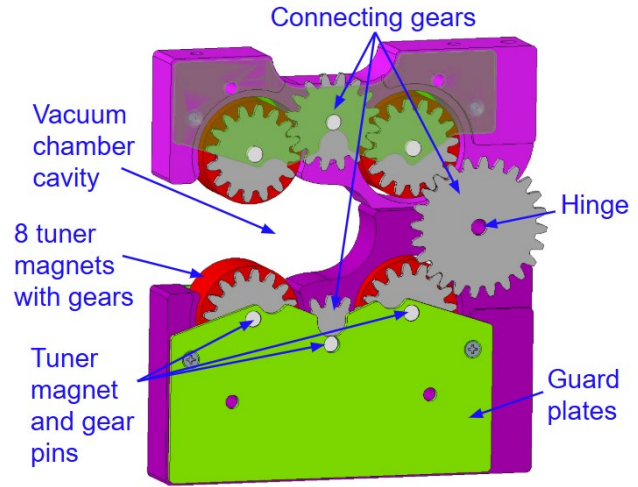


Figure 5: CAD model of half a tuner assembly showing functional features.

The functional features of half a tuner assembly are shown on Fig. 5. The other half is a mirror image. The guard plates keep the magnets secured, while the gears allow the magnets to rotate simultaneously with one another.

A hinge feature (Figs. 5 and 6) allows for the combined CT assembly to be opened and closed as one, as well as for inserting and removing from a vacuum chamber.

The corrector assembly is a set of fourteen diametrically magnetized neodymium rare earth permanent magnets that can be configured into a prescribed orientation to compensate small field errors.

The functional features of the corrector assembly are shown on Fig. 6. The magnet pins allow to adjust the orientation of each magnet, while the set screws push the magnets against flats, locking them into that orientation. Guard plates keep the magnets from pushing out.

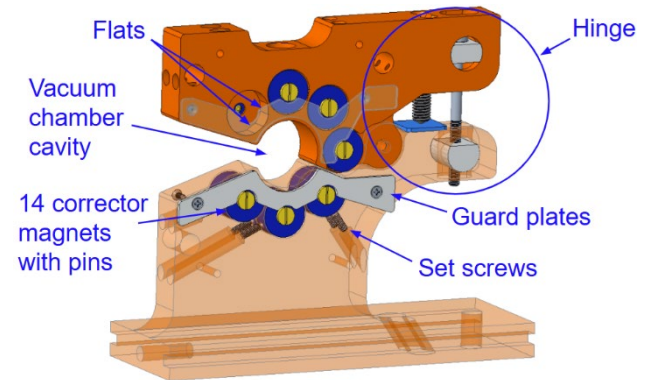


Figure 6: CAD model of the corrector assembly showing functional features.

Fabrication

Figure 7 shows both the tuner and corrector magnet holders assembled, where the corrector is sandwiched between two tuner halves. The crown gear allows for simultaneous rotation of magnets on both tuner halves.

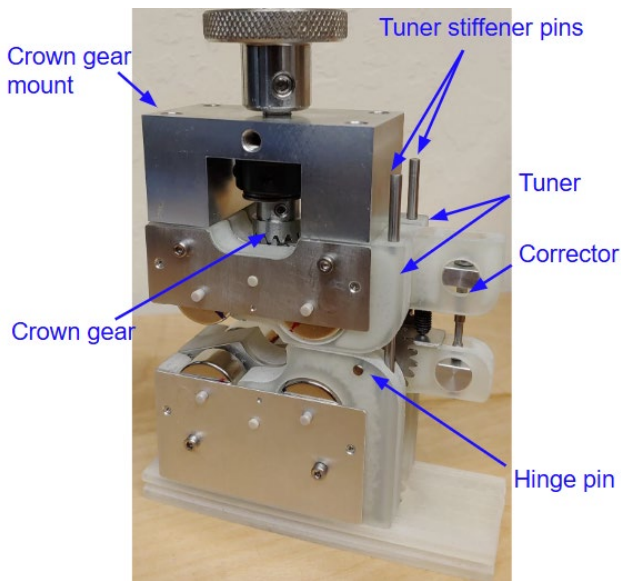


Figure 7: CT assembly.

The tuner and corrector holders are 3D printed from VisiJet M3 Crystal, followed by post machining. Both the crown gear and hinge pin are 316 SS, machined on a CNC and lathe, respectively. The tuner magnet and gear pins, as well as the corrector pins are machined from Polyether Ether Ketone (PEEK). The Tuner stiffener pins are 18-8 SS, while all other parts on the CT assembly are machined out of aluminium.

Magnetic Performance

Figure 8 shows the field orientation for the four magnets on each tuner half, where both tuner halves would be in this orientation together to get the full strength quadrupole. The quadrupole field strength can then be modified by using the crown gear to rotate the magnets on each tuner half in opposite directions from this nominal state.

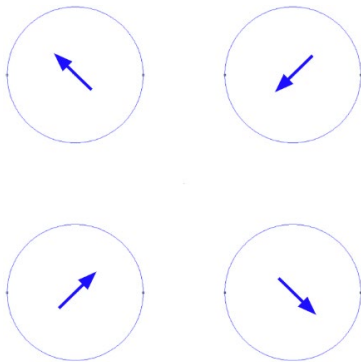


Figure 8: Quadrupole magnetic arrangement for half a tuner assembly.

Figure 9 below shows the measured quadrupole component of a complete tuner assembly (all eight magnets),

measured on a stretched wire measurement system. The magnets were rotated 360 degrees in 15 degree increments and the quadrupole component measured ± 1 T/m, allowing for in situ quadrupole focusing adjustments.

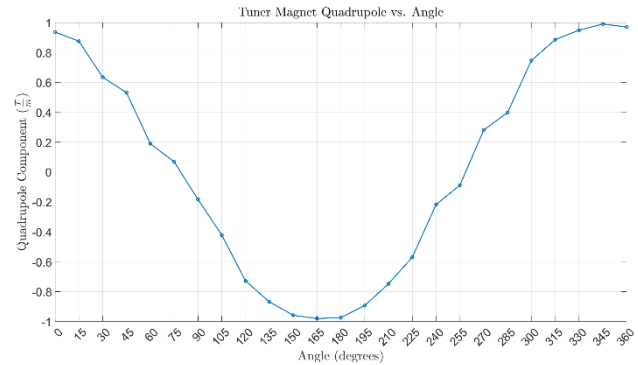


Figure 9: Tuner assembly quadrupole component measured against tuner magnet orientation.

CONCLUSION

This paper describes the mechanical design, fabrication and testing of two permanent magnet assemblies. The first is a dipole-quadrupole magnet assembly, providing a 1 T dipole + 50 T/m quadrupole field aimed to steer electrons. The second is an assembly to adjust a set of tuner and corrector permanent magnets. Regardless of the magnetic forces involved, the tuner magnets can rotate simultaneously and provide a ± 1 T/m quadrupole field for in situ quadrupole focusing strength adjustments, while corrector magnets can be oriented into a prescribed configuration to compensate for small field errors.

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